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AutoTurb: Using large language models for automatic algebraic turbulence model discovery

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ABSTRACT

Symbolic regression (SR) methods have been extensively investigated to explore explicit algebraic Reynolds stress models (EARS) for turbulence closure of Reynolds-averaged Navier-Stokes (RANS) equations. The deduced EARS can be readily implemented in existing computational fluid dynamic (CFD) codes and promotes the identification of physically interpretable turbulence models. Recently, large language models (LLMs) trained on large amounts of publicly available source code have drawn great attention for their abilities to automatically discover equations with more general free-text inputs and problem descriptions and provide wider possibilities with novel insights. This work proposes a novel framework, named “AutoTurb,” using LLMs to automatically discover algebraic expressions for correcting the linear Reynolds stress model. The direct Reynolds stress and the indirect RANS output (e.g., velocity field) are both involved in the training objective to guarantee data consistency and avoid numerical stiffness. An evolutionary search framework is used for global optimization, where constraints on functional complexity and simulation convergence are integrated into the objective to manage the extensive flexibility of LLMs. The proposed method is performed for separated flow over periodic hills. The generalizability of the discovered model is verified on a set of 2D turbulent separated flows with different Reynolds numbers and geometries. Results show that the corrected RANS enhances predictions of both Reynolds stress and mean velocity fields. Compared to models from other studies, our discovered model shows superior accuracy and generalization capability. The proposed approach provides a promising paradigm for using LLMs to improve turbulence modeling for a given class of flows.

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I. INTRODUCTION

Computational fluid dynamics (CFD) methods, including Reynolds-averaged Navier-Stokes (RANS),¹ large-eddy simulation (LES),² and direct numerical simulations (DNS),³ have been widely used to model and study fluid mechanisms⁴ over the past decades. Among them, the RANS method has gained widespread applications in industrial fields such as aerospace engineering,^{5,6} due to its lower computational cost and relatively reliable accuracy. However, for flow with separations, strong pressure gradients, and curvature, the RANS method frequently fails to deliver satisfactory simulation outcomes due to the linear eddy-viscosity assumption and semi-empirical turbulence modeling.^{7–9} Therefore, improving turbulence modeling in RANS to achieve accurate predictions of complex flow behaviors is crucial for engineering applications.

In recent years, thanks to the rapid development of high-performance computing architectures as well as increasing accessibility of high-fidelity flow data, data-driven approaches leveraging machine

learning (ML) techniques have emerged as promising techniques for improving the performance of RANS¹⁰ by correcting the Reynolds stress^{11,12} or turbulence closure models.^{13–15} These technologies provide novel paradigms for analyzing and understanding turbulent flows. However, the black-box nature of most ML methods, such as random forests¹⁶ or neural networks,^{12,17} makes it difficult to ensure stable simulation convergence and hinders the ability to provide physical interpretations due to the absence of explicit algebraic expressions. Research has demonstrated that when correction models are trained offline using high-fidelity Reynolds stress fields, the modified RANS equations can be ill-conditioned.¹⁸ Even when Reynolds stresses with errors less than 0.5% of the DNS database are substituted into RANS equations, the resulting velocity fields can exhibit significant errors (up to 35%).¹⁹ This limitation impedes their generalization to other flow scenarios and the inference of new models. The lack of stability, interpretability, and generalizability limits the use of these models within the engineering community and increases the difficulty of disseminating

learned models to end users due to the absence of explicit mathematical equations.

In order to avoid the aforementioned problems associated with black-box ML models, symbolic regression (SR) approaches have been introduced to the automatic discovery of equations and gained intensive attention in the turbulence modeling community. The goal of symbolic regression is to find explicit algebraic expressions that best predict the target given input variables. Once derived, the expressions can operate independently of the training environment and seamlessly integrate into existing CFD solvers with a negligible increase in computational cost. Besides, researchers can incorporate relevant functional forms and physical quantities into the training process based on turbulence laws and physical insights, ensuring that the resulting model possesses greater physical significance.

Depending on the search method used, SR methods used in turbulence modeling can be classified into four categories: 1) the gene expression programming (GEP) method: First introduced by Weatheritt and Sandberg,²⁰ this approach uses an evolutionary algorithm (EA) to find the global optimal tensor regression for modeling the anisotropy of Reynolds stress. Candidate solutions are encoded as linear strings of predefined operators and variables. A CFD-driven approach was further developed to use any flow feature from CFD results for model training.²¹ Then, a multi-objective evolutionary algorithm was proposed to combine multiple training objectives.²² 2) The SpARTA method, proposed by Schmelzer *et al.*,²³ implements a sparse regression with regularization terms to model corrections to Reynolds stress anisotropy and the production of transported turbulent quantities. This approach requires a predefined library of candidate expressions and operators, with the target corrections resolved using a frozen-RANS framework. 3) A CFD-driven symbolic identification method (CFD-driven SpARTA) is further developed,²⁴ which uses a surrogate-based optimization method to derive the optimal expression. 4) The deep SR (DSR) method, proposed by Tang *et al.*,²⁵ combines deep learning with SR. Using a risk-seeking policy gradient algorithm, neural networks are trained to generate the best expression for the Reynolds stress model (RSM). Besides the search method, the main characteristics of these SR methods are summarized in Table I. It shows that predefined operators or functional candidates are required by the existing methods. Managing functional complexity is crucial in SR methods to obtain pragmatic and interpretable models. In GEP, the length and number of genes in the encoded solution are defined to restrict the complexity. In SpARTA and CFD-driven SpARTA methods, Lasso and Ridge regressions are used to promote sparsity and constrain coefficients to relatively small values. For DSR, the minimal and maximal lengths of expressions are pre-specified and handled in the

prior sampling stage. Current methods place a significant burden on researchers, requiring extensive expertise and considerable human effort to design the algorithm framework.

In recent years, large language models (LLMs),²⁶ which are trained on large amounts of publicly available source codes, have become increasingly powerful due to intensive training data and large-scale model architectures. Pre-trained LLMs show impressive capabilities in language processing and code generation. It has drawn great attention in areas such as algorithm design,^{27,28} mathematical discovery,²⁹ and optimization.^{30–32} Recent work has shown that in comparison to standalone LLMs, incorporating LLMs in a search framework could significantly boost the performance in tackling some hard reasoning and designing tasks.^{28,33} The application of LLMs in scientific discovery offers a versatile and automated approach, considerably reducing the need for human intervention and specialized expertise during the design process.

In this paper, we present the first attempt to adopt LLMs for turbulence model discovery of the RANS, named AutoTurb, as shown in Table I. LLMs are used to formulate the optimization problem using natural language and autonomously generate algebraic expressions for correction models. To maintain data consistency and avoid numerical instability, a CFD-driven approach is used, with constraints placed on the complexity of the generated expressions. Moreover, the vast design freedom provided by LLMs necessitates strict constraints on numerical convergence in CFD to ensure a stable optimization process. Applications in different numerical experiments, including periodic hills with different geometry and Reynolds numbers, converging-diverging channels, and a curved backward-facing step, were used to validate the generability of the proposed method.

The structure of this paper is organized as follows: In Sec. II, we introduce the governing equations of RANS and the mathematical formulation of the k - ω shear stress transport (k - ω SST) turbulence model. Additionally, it outlines the basic assumptions of the nonlinear Reynolds stress model used in this paper. Section III presents the proposed framework based on LLMs. Numerical experiments for training and validating are presented in Secs. IV and V. Section VI concludes this paper and outlines future research directions.

II. PRELIMINARY

This section briefly recalls preliminary methodologies, including governing equations of incompressible RANS, the baseline linear viscosity model combined with the k - ω SST turbulence model, and the foundations of nonlinear algebraic correction for RSM.

TABLE I. Key features of existing SR methods and our approach.

Methods	Expression	Search method	Constraints
GEP ²⁰	Chromosome encoded as a linear string	Evolutionary algorithms (EA)	...
SpARTA ²³	Library of candidates with coefficients	Linear regression	Complexity & robustness
CFD-driven SpARTA ²⁴	Library of candidates with coefficients	Surrogate-based method	Complexity & robustness
DSR ²⁵	Sampled token of operator and variables	Gradient-based method	...
AutoTurb (Ours)	Arbitrary form and length	LLMs + EA	Complexity & convergence

A. RANS with turbulence modeling

The steady incompressible RANS equations are

$$\frac{\partial U_i}{\partial x_i} = 0, \quad (1)$$

$$U_j \frac{\partial U_i}{\partial x_j} = \frac{\partial}{\partial x_j} \left[-\frac{P}{\rho} + \nu \frac{\partial U_i}{\partial x_j} + \tau_{ij} \right],$$

where U_i with $i \in \{1, 2, 3\}$ and P are the components of the mean flow velocity and the mean pressure. ρ and ν are the constant density and the kinematic viscosity. The instantaneous fluctuation of turbulence is represented by a Reynolds stress tensor $\tau_{ij} = -\langle u_i u_j \rangle$ in the momentum equation. Known as the Boussinesq assumption, the anisotropic part a_{ij}^B of Reynolds stress can be modeled by a linear constitutive relationship with the mean-flow straining field.

$$a_{ij}^B = \tau_{ij} + \frac{2}{3} \rho k \delta_{ij} = 2\nu_t S_{ij}, \quad (2)$$

where ν_t is the turbulent kinematic viscosity or eddy viscosity, $k := \frac{1}{2} \tau_{ii}$ is the turbulence kinetic energy (TKE), and S_{ij} is the traceless mean strain rate tensor, which can be written as $S_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right)$ for an incompressible flow.

The eddy viscosity ν_t can be computed from two transported variables, such as k and the specific turbulence dissipation rate ω . In this work, the popular k - ω SST model is adopted as the baseline turbulence model, which defines the transport equations for k and ω as follows:

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = \underbrace{\tau_{ij} \frac{\partial U_i}{\partial x_j}}_{P_k} - \beta^* k \omega + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial x_j} \right], \quad (3)$$

$$\frac{\partial \omega}{\partial t} + U_j \frac{\partial \omega}{\partial x_j} = \frac{\gamma}{\nu_t} P_k - \beta \omega^2 + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_\omega \nu_t) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \sigma_{\omega 2} \frac{1}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i}. \quad (4)$$

Then, the eddy viscosity ν_t is modeled as

$$\nu_t := \frac{a_1 k}{\max(a_1 \omega, SF_2)}. \quad (5)$$

All remaining terms and coefficients are omitted for brevity; see Ref. 34 for details. The k - ω SST model takes both the wall treatment and free-stream turbulence properties into account. It usually demonstrates good performance in flows with adverse pressure gradients and separation. However, because of the linear eddy-viscosity assumption and TKE equilibrium assumption in the boundary layer, the k - ω SST model often yields an unsatisfactory prediction of flow separation and tends to underestimate Reynolds stress.

B. Nonlinear algebraic correction for Reynolds stress model

The correction of the linear constitutive relationship for RSM is implemented based on an explicit nonlinear framework proposed by Pope.³⁵ It is assumed that RSM depends not only on the strain rate

tensor S_{ij} but also on the rotation rate tensor $\Omega_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i} \right)$. Then, the most general form of the correction for non-dimensional RSM anisotropy is derived as a polynomial of 10 basis tensors and five Galilean invariants

$$\frac{a_{ij}^A}{2k} = b_{ij}^A(\tilde{S}_{ij}, \tilde{\Omega}_{ij}) = \sum_{n=1}^{10} T_{ij}^{(n)} \alpha_n(I_1, \dots, I_5), \quad (6)$$

where $T_{ij}^{(n)}$ are basis tensors and $\alpha_n(\cdot)$ is an arbitrary scalar function of invariants I_m , $m = 1, \dots, 5$. The invariants I_m are expressed as functions of the non-dimensional $\tilde{S}_{ij}$ and $\tilde{\Omega}_{ij}$ that are both normalized by the time scale $1/\omega$.

For 2D flows, we only consider the first three base tensors and the first two nonzero invariants, which are given by

$$\begin{aligned} T_{ij}^{(1)} &= \tilde{S}_{ij}, & I_1 &= \tilde{S}_{ij} \tilde{S}_{ji}, \\ T_{ij}^{(2)} &= \tilde{S}_{ik} \tilde{\Omega}_{kj} - \tilde{\Omega}_{ik} \tilde{S}_{kj}, & I_2 &= \tilde{\Omega}_{ij} \tilde{\Omega}_{ji}, \\ T_{ij}^{(3)} &= \tilde{S}_{ik} \tilde{S}_{kj} - \frac{1}{3} \delta_{ij} \tilde{S}_{mn} \tilde{S}_{nm}. \end{aligned} \quad (7)$$

Apart from the correction on RSM anisotropy, an additional correction R is imposed on the production term in the k - ω SST transport equations to correct the TKE equilibrium assumption. Following the work in Refs. 23 and 24, the correction term R_{ij} takes the form

$$R_{ij} = 2kb_{ij}^R \frac{\partial U_i}{\partial x_j}, \quad (8)$$

and the correction tensor b_{ij}^R is modeled in the same way of b_{ij}^A as follows:

$$b_{ij}^R = \sum_{n=1}^{10} T_{ij}^{(n)} \alpha_n^R(I_1, \dots, I_5). \quad (9)$$

In the following, we developed an LLMs-based symbolic regression method to discover the optimal algebraic expressions of b_{ij}^A and b_{ij}^R to correct the turbulence model using high-fidelity data.

III. PROPOSED METHOD: AUTOTURB

This work targets using LLMs to automatically design algebraic correction terms for the turbulence model in the RANS solver to make it a better approximation of expensive high-fidelity experimental results. The proposed framework using LLMs, named “AutoTurb,” is given in Sec. III A. The formulation of fitness for evaluating the discovered models and the details of the LLMs-based search method are presented in Secs. III B and III C.

A. AutoTurb framework

The AutoTurb framework consists of three blocks, as illustrated in Fig. 1: 1) the LLMs-based search block, 2) the prompt block, and 3) the CFD simulation block.

- (1) The LLM-based search block is the high-level block that iteratively prompts the LLMs to generate new algebraic expressions and invokes the simulation block to evaluate their fitness value. It uses an evolutionary algorithm to guide LLMs in producing new expressions to recover the target fields. Each individual in

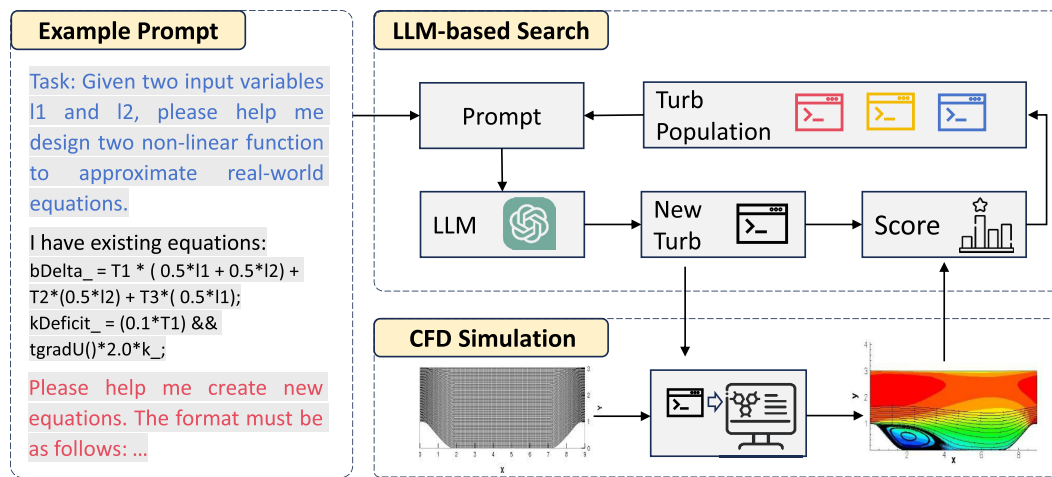


FIG. 1. AutoTurb framework.

the population represents a unique turbulence model, characterized by two distinct algebraic correction terms (b_{ij}^A, b_{ij}^R).

- (2) The prompt block generates new algebraic expressions for the two correction terms based on prompt strategies. It selects parent expressions from the population and utilizes LLMs to create offspring expressions, fostering exploration and diversity in the evolutionary search.
- (3) CFD simulation block: Each new pair of formulated algebraic correction expressions gives rise to a novel turbulence model, which is subsequently integrated into the RANS solver. By compiling a RANS solver with the new turbulence model, a simulation is performed over the training set. The performance of the new turbulence model is quantified by the discrepancy between the simulation results and high-fidelity experimental data, which serves as the objective value for scoring during the search process.

AutoTurb leverages the strengths of LLMs in generating complex expressions and the robustness of the evolutionary framework to iteratively improve the fitness of generated models. The integration of LLMs into the evolutionary search framework primarily serves to generate a diverse set of new expressions. While LLMs can produce both superior and inferior expressions in each iteration, the selection and survival of these expressions are governed by the evolutionary principles of mutation, crossover, and selection based on fitness evaluations. The objective function plays a critical role here, as it quantitatively assesses the fitness of each expression, ensuring that elite individuals contribute to the next generation. This process is akin to natural selection, where the “fittest” individuals are chosen to reproduce. Theoretical results from evolutionary algorithms support this approach, showing that such mechanisms can effectively explore and exploit the search space, leading to the gradual improvement of solutions over generations. This is underpinned by the schema theorem and convergence theories, which suggest that good building blocks of solutions are preserved and propagated through generations.³⁶

Integrating LLMs into the evolutionary framework offers significant advantages, particularly in terms of automation and flexibility.

Unlike traditional methods such as GEP, where the preliminary set of functions and operators must be meticulously defined and tailored to the specific problem domain, LLMs can generate a wide variety of expression structures dynamically. This capability reduces the need for extensive domain-specific preparations and allows the system to adapt more freely to the nuances of the task at hand. Furthermore, LLMs can leverage vast amounts of data and complex patterns that might be infeasible for humans to encode manually, thereby enhancing the innovative potential of the generated solutions. This automation and adaptability make the proposed framework not only efficient but also broadly applicable across different expression optimization tasks.

B. Objective formulation

In this work, both the internal Reynolds stress and the external mean velocity field by RANS simulation are utilized as target quantity fields to train the models. To maintain data consistency and avoid numerical instability, a CFD-driven approach is used, with constraints placed on the complexity of the generated expressions. Moreover, the extensive design flexibility provided by LLMs necessitates strict constraints on numerical convergence in RANS simulation. These constraints are essential to prevent divergence or ill-conditioning, ensuring a stable and reliable optimization process.

Specifically, the following metrics are considered in the objective:

- Minimizing the discrepancy between the target mean velocity field and the observed velocity field by running the RANS solver involves implementing a corrective turbulence model.

$$f_1 = \|U^* - U(b_{ij}^A, b_{ij}^R)\|_2^2.$$

- Minimizing the discrepancy between the target TKE field and the observation k by solving the turbulence transport equations

$$f_2 = \|k^* - k(b_{ij}^A, b_{ij}^R)\|_2^2.$$

- Reducing the complexity of the expressions regarding the number of functional operators n_o . We assume that simple expressions

with $n_o \leq 10$ are preferred. To achieve this, a segmented function is given, as shown in Fig. 2.

$$f_3 = \begin{cases} \sqrt{n_o + 1000}/\sqrt{1001}, & \text{if } n_o \leq 10, \\ \sqrt{(n_o)^2 + 1000 - 90}/\sqrt{1001}, & \text{if } n_o > 10. \end{cases}$$

- Ensuring the numerical convergence of the simulation by restricting the residual of the resolved mean pressure $Res(P)$ to an acceptable threshold 10^{-6}

$$f_4 = \begin{cases} 1, & \text{if } Res \leq 10^{-6}, \\ Res/10^{-6}, & \text{if } Res > 10^{-6}. \end{cases}$$

The first two metrics are computed and averaged over all mesh points of the training data set, where the superscript $*$ denotes quantities evaluated from the high-fidelity data. The last two merits or constraints are introduced to reduce model complexity and avoid overfitting and divergence, as LLMs have the capability of producing expressions of arbitrary form and length. The final objective is formulated as

$$f = (f_1 + 0.5 \times f_2) \times f_3 \times f_4. \quad (10)$$

C. LLM-based search method

The LLM-based search method uses an evolutionary algorithm to iteratively prompt LLMs to generate new expressions to explore novel and elite designs continuously. Two symbolic regression expressions for correcting the RSM and production term in $k-\omega$ SST are trained simultaneously using the proposed algorithm. We maintain a population of N pairs of expressions, denoted as $P = \{e_1, \dots, e_N\}$, at each generation. Each pair of expressions e_i corresponds to a distinct turbulence model. The model will undergo evaluation in the RANS solver and is assigned an objective value $f(e_i)$.

The LLMs-based search method is outlined as follows:

Step 0 Initialization: The initial population P of N pairs of expressions $\{e_1, \dots, e_N\}$ is generated by LLMs using the *Initialization Prompt*, illustrated in Appendix A.

Step 1 Model Search: If the stopping condition is not met, four *Evolution Prompt* strategies with diverse tradeoffs between exploration and exploitation are simultaneously used and repeated N times to generate $4N$ new algebraic expression pairs in one

generation. Details of these prompt strategies are provided in Appendix A.

For each prompt strategy, the following process is executed:

- Step 1.1: Select parent expressions from the current population to construct a prompt for the strategy.
- Step 1.2: Request LLMs to generate a new design of an expression pair along with its corresponding code implementation.
- Step 1.3: Incorporate the resulting new turbulence model into the RANS solver and simulate to evaluate its objective value.
- Step 1.4: Add the new model to the current population if both the model and code are feasible.

Step 2 Population Management: The top N individuals in terms of objective value from the current population are selected to form the population for the next generation. The process then returns to Step 1.

IV. NUMERICAL EXPERIMENTS SETTINGS

A. Experimental settings of AutoTurb

In our experiments, the pre-trained GPT-3.5-turbo LLM is used. We configure the LLM with a temperature setting of 1.0 and a top-p value of 1.0 to ensure diversity and randomness in generated expressions. During the search process, the number of generations in AutoTurb is set to 20, the population size N is 20, and the four prompt strategies are used simultaneously, which results in 1600 evaluations. For evaluation, the maximum number of RANS iterations is set to 10 000, and the maximum running time for each instance is 2000 s. The entire framework, including the AutoTurb workflow and its integration with the RANS solver, is implemented in Python, utilizing a Python wrapper for OpenFOAM. The experiments were executed on a single Intel Core i7-9700 CPU.

B. Datasets

This section presents the details of high-fidelity datasets used for training and cross-validating the models. We use the high-fidelity data of widely studied separated flow over periodic hills at $Re = 10\,595$ to train the symbolic regression model in the AutoTurb framework. Then, flows over periodic hills with varying steepness ratios at $Re = 5600$, a converging-diverging channel at $Re = 12\,600$, and a curved backward-facing step at $Re = 13\,700$ is used to demonstrate the performance of the trained model and compare it with models found by other SR methods. These datasets encompass diverse flow configurations, providing a reliable assessment of the generalizability and effectiveness of the trained model.

1. Train cases

The highly-resolved LES data of periodic hills at $Re = 10\,595$ from ERCOFTAC Database³⁷ Classic Collection case 81 is used for training the model, which is shortened as “PH.” The height, spanwise width, and periodic streamwise length of the computational domain are $(9H, 3.035H, 4.5H)$ in terms of the hill height H , as shown in Fig. 3. Cyclic boundary conditions are imposed at the hill crests. The top and bottom walls are treated as no-slip. A constant bulk velocity is maintained at the inlet by adding a pressure gradient source term. The Reynolds number 10 595 is based on the bulk velocity and the hill height.

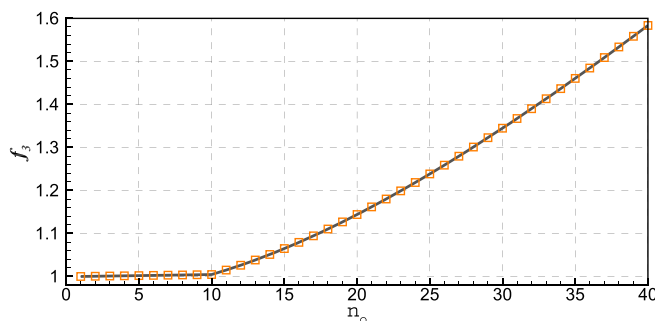


FIG. 2. Variation of f_3 vs the number of operators.

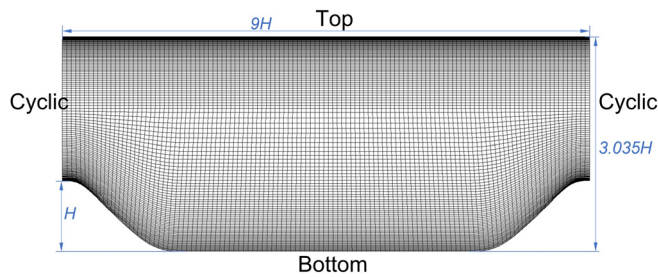


FIG. 3. Geometry and computational grid for periodic hills (PH) case at $Re = 10\,595$.

The RANS simulation utilizes a computational grid with 120×130 cells in the stream-wise and wall-normal directions for spatial discretization. The simulation was conducted using the open-source finite-volume solver OpenFOAM. Referred to Ref. 38, the same numerical schemes are consistently applied across all RANS simulations in this study. The SimpleFoam solver, which uses the semi-implicit method for pressure-linked equations-consistent (SIMPLEC) algorithm, is used to simulate the 2D incompressible, viscous, and steady flows. For turbulence modeling, the $k-\omega$ SST model is adopted as the baseline turbulence closure model. The numerical schemes include a second-order upwind scheme for the convective terms in the momentum equations, a first-order upwind scheme for the convective terms in the turbulence transport equations, and a second-order central difference scheme for the diffusive terms. The generalized geometric algebraic multigrid (GAMG) solver and the preconditioned bi-conjugate gradient (PBiCG) solver are used to solve the pressure equation and all other equations.

2. Cross-validation cases

A set of separated flows with varying geometries and Reynolds numbers are used to validate the generalization capability of the trained models, as shown in Table II.

- (1) Periodic hills. Four cases of periodic hills with steepness ratios $\{\alpha = 0.8, 1.0, 1.2, 1.5\}$ at $Re = 5600$ are used for validation. For each case, a computational grid consisting of 99×149 cells in stream-wise and wall-vertical directions is used for RANS simulation. An illustration of the case with steepness $\alpha = 0.8$ is shown in Fig. 4. DNS simulation results are provided by Xiao *et al.*³⁹ using the Incompact3d solver and interpolated to RANS

TABLE II. Cross-validation test cases.

Flow cases	Notation	Re	No. of grids	Ref
Periodic hills with different steepness	$PH\alpha$	5600	99×149	39
Converging-diverging channel	CD	12 600	140×150	40
Curved backward-facing step	CBF	13 700	140×180	41

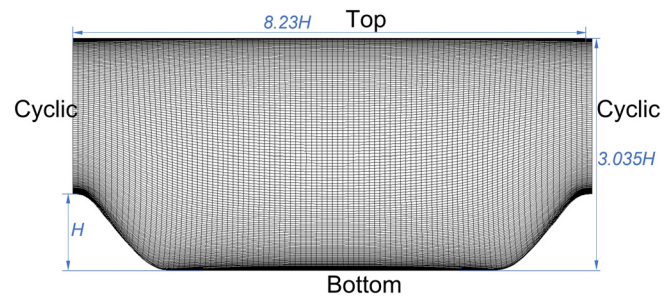


FIG. 4. Geometry and computational grid for periodic hills case with $\alpha = 0.8$ ($PH_{\alpha=0.8}$), $Re = 5600$.

grids for comparison. It is shown that flows separate after the first hill crest. As the hills become flatter, the separation region decreases in size. The separation bubbles are overestimated by the baseline $k-\omega$ SST closure model. The case with $\alpha = 0.5$ is not included in this work, as the baseline model can provide simulation results with acceptable accuracy for this configuration. The remaining cases are abbreviated as “ $PH_{\alpha_{0.8}}$,” “ $PH_{\alpha_{1.0}}$,” “ $PH_{\alpha_{1.2}}$,” and “ $PH_{\alpha_{1.5}}$.”

- (2) Converging-diverging channel. The second case, labeled as “CD,” is a channel flow with an asymmetric hill that has a height of $2/3H$, located near $x = 5.215\,64$. The Reynolds number for this flow, based on the channel half-height and the maximum velocity U_{max} at the inlet, is 12 600. A fully developed channel flow at the same Reynolds number is applied as the inlet condition. High-fidelity DNS data from Laval and Marquillie⁴⁰ indicates the presence of a small separation region on the lee side of the hill crest. For RANS simulations, a computational grid with 140×150 cells is used. The computational grid and geometry of this case are shown in Fig. 5.
- (3) Curved backward-facing step. The third test case, labeled as “CBF,” involves a 2D flow over a gently curved backward-facing step with a height of H . The upstream channel height is $8.52H$ and the Reynolds number, based on U_{max} at the inlet and step height, is 13 700. A fully developed channel flow at the same Re serves as the inlet condition. High-fidelity LES data from Bentalb *et al.*⁴¹ is used for training the model. The RANS simulations are performed on a computational grid of 140×180 cells, depicted in Fig. 6.

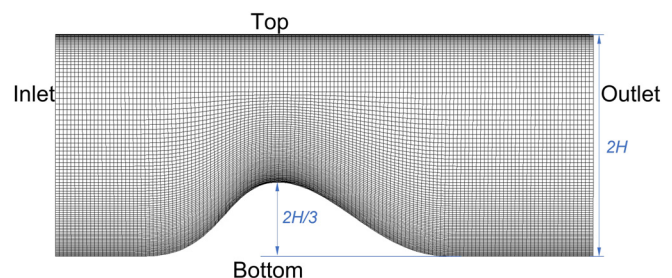


FIG. 5. Geometry and computational grid for converging and diverging channel (CD) case, $Re = 12\,600$.

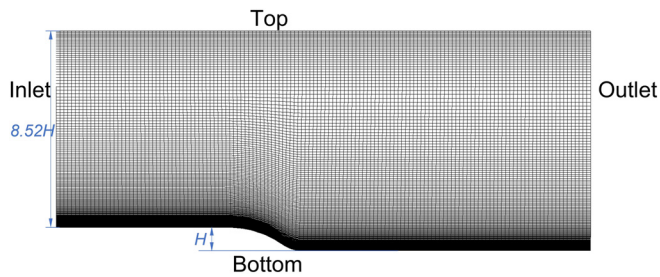


FIG. 6. Geometry and computational grid for curved backward-facing step (CBF) case, $Re = 13\,700$.

V. RESULTS

A. Model discovery

The convergence curves of the observed best objective for the LLMs-based evolutionary search are depicted in Fig. 7. The y-axis represents the custom objective value, while the x-axis denotes the generation number. Each blue solid circle corresponds to a turbulence model designed by LLMs during the evolutionary process. The red line highlights the best-performing model in each population as the evolution progresses. A clear convergence is observed within 20 generations, with the best objective value decreasing from 0.0023 to 0.0017. Initially, two models in the population failed to converge in the simulation, exhibiting a large residual error (these are not shown in the convergence curve due to their extremely high objective values). In contrast, after 20 generations of evolution, all models in the final population demonstrate reduced complexity and enhanced robustness during simulations.

The optimum algebraic expressions for the two corrective terms found by AutoTurb are:

$$\text{Model - LLMs} : \begin{cases} b_{ij}^{\Delta} = 0, \\ b_{ij}^R = (\sin(I_1) + 0.5)T_{ij}^1 + T_{ij}^2. \end{cases} \quad (11)$$

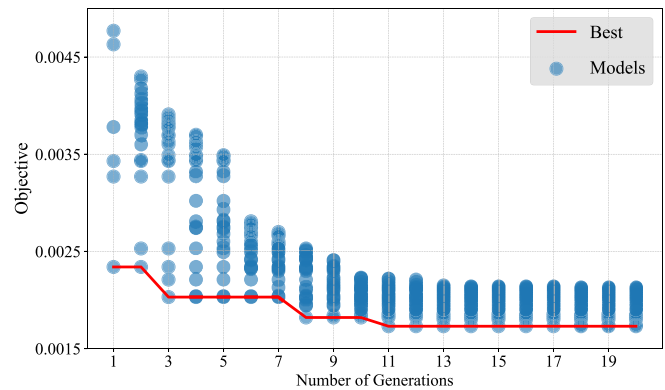


FIG. 7. Convergence curves of objective function (each blue sample formulates a turbulence model).

The resulting expressions maintain their elegant simplicity due to the complexity and convergence constraints imposed on the objective. This simplicity is crucial in avoiding overfitting and ensures robust performance across different scenarios. The observed velocity streamlines and TKE fields by RANS with the discovered model are shown in Fig. 8. Both the velocity and TKE fields are significantly improved.

In this particular training case, the linear eddy-viscosity constitutive model is preserved, since the correction term b_{ij}^{Δ} remains zero. However, a correction term R_{ij} is added to the production term P_k in the TKE transport equation, as depicted in Fig. 9. This adjustment significantly amplifies the magnitude of P_k , leading to a higher value of k . The correction increases both turbulence intensity and eddy viscosity ν_t , as shown in Fig. 10, which enhances the model's resistance to flow separation. With the increase of eddy viscosity ν_t , the shear stress intensifies, leading to a reduction in the size of the separation bubble.

In the following section, we systematically evaluate the performance of our model by incorporating it into the RANS solver to solve

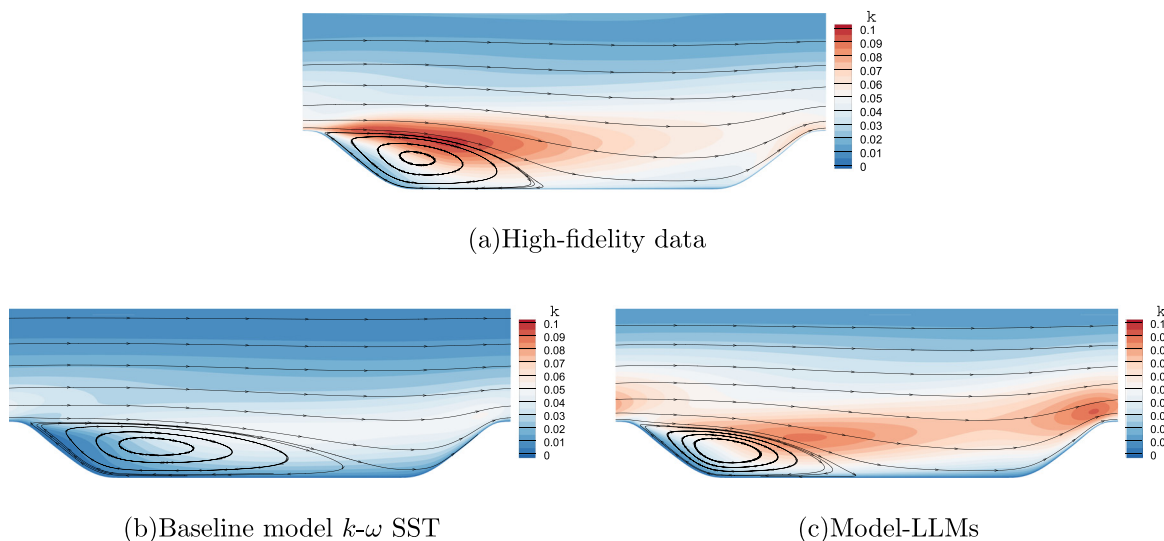


FIG. 8. Contours of TKE k with velocity streamlines.

FIG. 9. Correction to the production term P_k by model-LLMs.

flow fields and then compare the results with high-fidelity data. In addition, models identified by other SR methods are integrated into the RANS solver for comparison. Although plenty of SR methods have been developed, only a few models with explicit algebraic expressions of similar corrective terms for separated flows are available. Two models from Saidi *et al.*,²⁴ developed using the CFD-driven SpaRTA method, are included in the comparison. “Model 1” was trained against high-fidelity Reynolds stress data, and “Model 2” was trained using indirect data, including mean velocity and skin friction. A “Model frozen” reported by Schmelzer *et al.*,²³ which was trained using k -corrective frozen-RANS with the SpaRTA method, is also investigated. These models are presented in detail below:

$$\text{Model 1 : } \begin{cases} b_{ij}^A = (-0.147I_1^2)T_{ij}^1 + (-0.26791)T_{ij}^2, \\ b_{ij}^R = -0.46018T_{ij}^1 - 0.16779T_{ij}^3. \end{cases} \quad (12)$$

$$\text{Model 2 : } \begin{cases} b_{ij}^A = (-0.28356)T_{ij}^1 + (-0.14738I_2^2)T_{ij}^2, \\ b_{ij}^R = (-0.10375I_2 - 0.28833I_1I_2)T_{ij}^3. \end{cases} \quad (13)$$

$$\text{Model - frozen : } \begin{cases} b_{ij}^A = 0, \\ b_{ij}^R = -0.39T_{ij}^1. \end{cases} \quad (14)$$

B. Model evaluations

The four models, Model-LLMs, Model 1, Model 2, and Model-frozen, are all trained by the high-fidelity data of flow over periodic hills at $Re = 10595$. Mean squared errors (MSE) of the predictions for the mean velocity field U , turbulent kinetic energy k , and Reynolds shear stress anisotropy τ_{12} are shown in Table III. The profiles at $x/H = \{0.05, 1, 2, 3, 4, 5, 6, 7, 8\}$ are shown in Fig. 11. The corrected turbulence closure in this work greatly improves the mean velocity field compared with the baseline model. The model accurately captures a smaller separation region, aligning well with the high-fidelity LES data. Meanwhile, the discovered model provides a more accurate prediction for the Reynolds stress, which is proven by the improved

TABLE III. MSE of the prediction for training data using various SR models (PH, $Re = 10595$).

Models	U	k	τ_{12}
k - ω SST	4.39013×10^{-3}	3.90736×10^{-4}	3.69883×10^{-5}
Model-LLMs	6.22571×10^{-4}	1.18714×10^{-4}	2.05218×10^{-5}
Model 1	6.90418×10^{-4}	1.10777×10^{-4}	2.03590×10^{-5}
Model 2	7.64242×10^{-4}	1.88782×10^{-4}	2.68433×10^{-5}
Model-frozen	8.07702×10^{-4}	1.12319×10^{-4}	2.03887×10^{-5}

prediction for k and τ_{12} in Figs. 11(b) and 11(c). The MSEs of mean velocity and TKE are reduced by 14.2% and 30.4%, respectively.

All four models greatly improve the RANS simulation results compared with the baseline. Among them, the present model achieves the most accurate prediction for the velocity field, while comparable results for k and τ_{12} are provided by Model-LLMs, Model 1, and Model-frozen. This is because more weight is being assigned to achieving an accurate velocity field compared to the TKE k . Model 2 performs worse for Reynolds stress, as it was trained using only indirect data.

To evaluate the performance of the proposed approach in handling sparse data, another algebraic model is trained using several selected data slices. Specifically, velocity and TKE profiles at $x/H = \{0.05, 0.5, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0\}$ are used for training, representing only 8.3% of the total mesh points. The resulting model writes $b_{ij}^A = (\sin(4I_1))T_{ij}^1 + I_1^2T_{ij}^2$; $b_{ij}^R = \cos(I_2)T_{ij}^3$. Incorporating this model into the RANS solver, the estimated MSEs of U , k , and τ_{12} over the whole flow field are 5.805×10^{-4} , 3.385×10^{-4} , and 2.357×10^{-5} . It indicates that the model retrieves a more accurate velocity field compared to Model-LLMs. However, the prediction for Reynolds stress is worse but still outperforms the baseline model. The reason is that the extracted data is very sparse, leading to a loss of TKE features. Nonetheless, most of the velocity information is preserved due to the inclusion of an extra slice $x/H = 0.5$ within the separation region.

C. Model generalization

The discovered model is further validated on a set of separated flows over different geometries and Reynolds numbers, as given in Table II in Sec. IV B 2. The MSEs of the mean velocity field U , turbulent kinetic energy k , and Reynolds shear stress anisotropy τ_{12} by the models for all the test cases are depicted in Fig. 12. The concrete MSEs are given in Table IV in Appendix B. In all cases, the errors are normalized with the MSE of the baseline k - ω SST model. The profiles of U , k , and τ_{12} at a sequence of streamwise locations for PH $\alpha_{0.8}$, PH $\alpha_{1.0}$, PH $\alpha_{1.2}$, PH $\alpha_{1.5}$, CD and CBF are displayed in Figs. 13–18 in

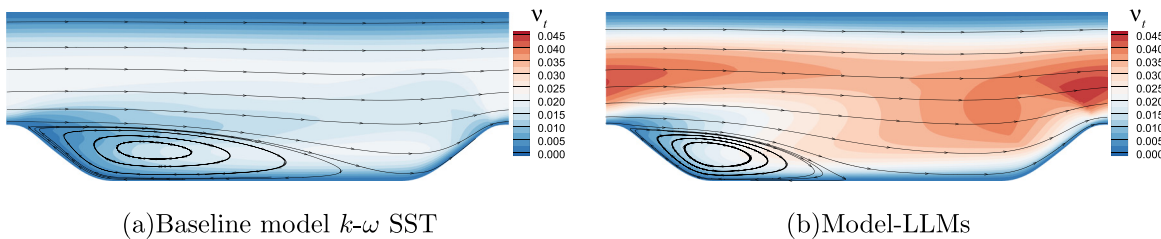


FIG. 10. Contours of eddy viscosity with velocity streamlines.

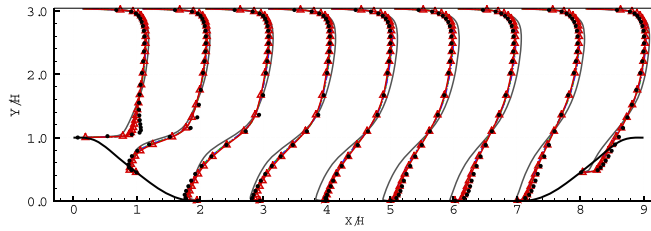
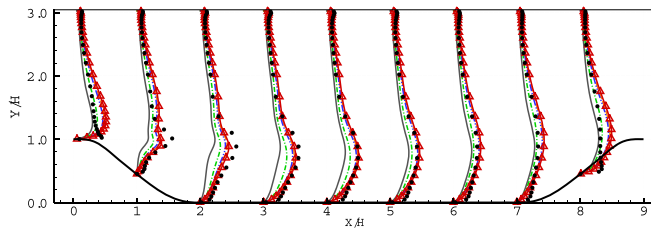
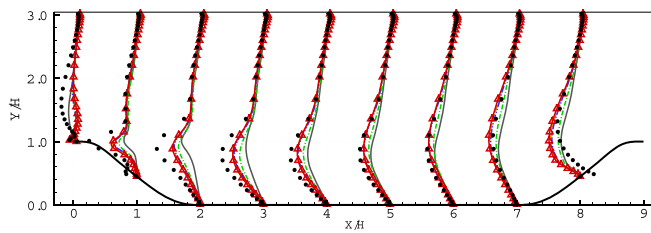
(a) Streamwise velocity profiles ($U_1/U_b + x/H$)(b) Turbulent kinetic energy ($6k/U_b^2 + x/H$)(c) Reynolds shear stress ($20\tau_{12}/U_b^2 + x/H$)

FIG. 11. Estimation for various SR models for flow over Periodic hills at $Re = 10595$. High-fidelity data: ($\cdots$), $k-\omega$ SST: ($—$), Model 1: ($- - -$), Model 2: ($- \cdot - \cdot -$), Model-frozen: ($- \cdot - \cdot - \cdot$), Model-LLMs: (inlinefigure).

Appendix C. The models significantly improve the simulation results compared to the baseline model. Significantly enhanced velocity fields are retrieved with more accurate separation regions across all cases. However, the improvement in the velocity field is relatively less in the $PH\alpha_{0.8}$ case, as the baseline model already exhibits a relatively low error in this case.

It is observed that the present model produces the smallest error for the velocity field in the training case PH, $PH\alpha_{1.2}$, and $PH\alpha_{1.5}$ at $Re = 5600$, and in the CD and CBF cases. Meanwhile, it provides the most accurate predictions for k in the training case PH, $PH\alpha_{1.2}$, $PH\alpha_{1.5}$, and CBF, and generates the best predictions for τ_{12} in all cases except for the $PH\alpha_{0.8}$ case. Overall, the present model demonstrates the best U and Reynold stress performance across most cases.

The accuracy of the current model is slightly better than Model 1 and model-frozen. Model 1, which is trained solely using the Reynolds stress, performs best in predicting Reynolds stress anisotropy for most cases. The present model achieves lower errors in velocity prediction

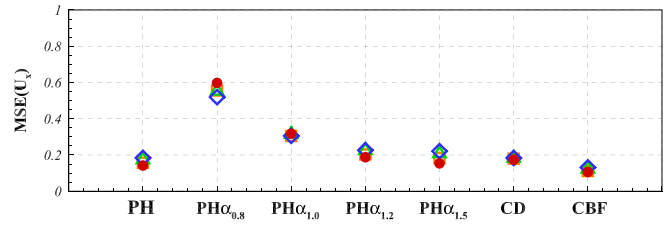
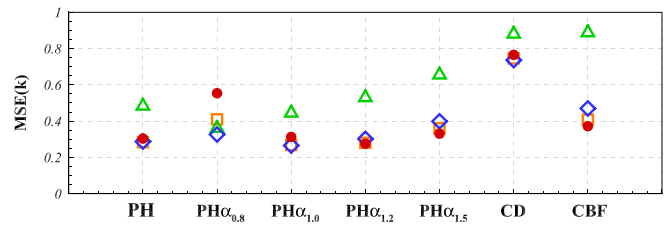
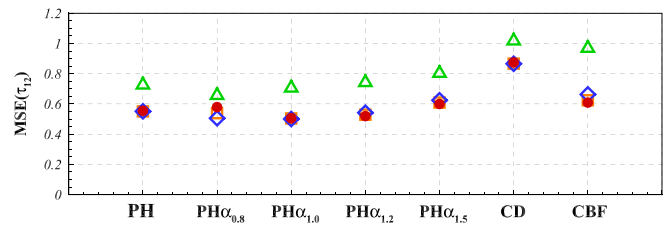
(a) Streamwise velocity profiles (U_1)(b) Turbulent kinetic energy (k)(c) Reynolds shear stress (τ_{12})

FIG. 12. Mean squared errors for flow quantities of cross-validation cases using various models. Model 1: ($\square$), Model 2: ($\triangle$), Model-frozen: ($\diamond$), Model-LLMs: ($\bullet$).

compared to the Model 1, as it is involved in the objective function. Model-frozen shows slightly worse performance than the current Model-LLMs, as it incorporates direct and indirect data in an implicit offline manner. On the contrary, Model 2, trained using only indirect data, cannot retrieve the Reynolds stress field as accurately as other models, though it still outperforms the baseline model.

VI. CONCLUSION

In this work, a novel framework, AutoTurb, is proposed, leveraging LLMs to automatically discover algebraic expressions for correcting turbulence closure models. This approach introduces two algebraic correction terms: one for the Reynolds stress model and another for the production term in the $k-\omega$ SST TKE transport equation. LLMs are utilized to prompt the optimization problem using natural language and autonomously generate algebraic expressions for the correction terms. The objective function is formulated by integrating direct observations of TKE and indirect velocity output from the CFD model to ensure data consistency. Additionally, constraints on functional complexity and numerical convergence of the corrected RANS solver

are imposed to prevent divergence or ill-conditioning, ensuring a stable optimization process.

The proposed method is performed for separated flow over periodic hills at $Re = 10\,595$. For the discovered model, the linear eddy-viscosity constitutive model is preserved, while the production term P_k is enlarged. Consequently, the eddy viscosity ν_t is increased, which enhances the model's resistance to flow separation, and a smaller separation bubble is observed. Compared with the baseline model, the discovered model provides more accurate predictions for the velocity field and the Reynolds stress.

TABLE IV. Mean-squared errors of the predictions for cross-validation cases using various SR models.

(a) Periodic hills $Re = 5600$ and $\alpha = 0.8$				
Models	Model-LLMs	Model 1	Model 2	Model-frozen
U	0.597 32	0.555 72	0.552 36	0.519 10
k	0.553 90	0.410 03	0.360 72	0.326 21
τ_{12}	0.578 99	0.538 99	0.656 63	0.505 48
(b) Periodic hills $Re = 5600$ and $\alpha = 1.0$				
Models	Model-LLMs	Model 1	Model 2	Model-frozen
U	0.317 49	0.304 96	0.314 16	0.305 59
k	0.312 81	0.268 16	0.444 71	0.265 29
τ_{12}	0.505 74	0.504 01	0.705 44	0.500 52
(c) Periodic hills $Re = 5600$ and $\alpha = 1.2$				
Models	Model-LLMs	Model 1	Model 2	Model-frozen
U	0.187 71	0.202 42	0.225 4	0.226 40
k	0.275 40	0.280 38	0.529 84	0.302 24
τ_{12}	0.519 52	0.528 75	0.742 05	0.541 79
(d) Periodic hills $Re = 5, 600, \alpha = 1.5$				
Models	Model-LLMs	Model 1	Model 2	Model-frozen
U	0.153 56	0.182 88	0.207 66	0.220 88
k	0.330 78	0.360 53	0.655 10	0.398 68
τ_{12}	0.599 56	0.609 11	0.803 35	0.624 24
(e) Converging-diverging channel $Re = 12\,600$				
Models	Model-LLMs	Model 1	Model 2	Model-frozen
U	0.174 62	0.179 86	0.182 74	0.183 34
k	0.765 40	0.745 75	0.880 18	0.736 50
τ_{12}	0.874 47	0.867 06	1.017 43	0.866 04
(f) Curved back-facing step $Re = 13\,700$				
Models	Model-LLMs	Model 1	Model 2	Model-frozen
U	0.106 12	0.111 26	0.122 71	0.131 53
k	0.371 98	0.408 03	0.887 75	0.469 85
τ_{12}	0.609 13	0.624 28	0.968 93	0.663 00

The generalizability of the discovered model is validated on a series of 2D turbulent separated flow configurations with varying Reynolds numbers and geometries and compared with other SR models. The present Model-LLMs exhibit the best overall performance in predicting both U_1 and Reynolds stress across most cases. It surpasses Model 1 in velocity prediction and outperforms Model 2 in predicting Reynolds stress quantities. Despite Model-frozen being trained with both direct and indirect data, it still falls short of the discovered model, which benefits from a CFD-driven training approach.

The proposed AutoTurb framework using LLMs provides a promising paradigm for using LLMs to improve turbulence modeling

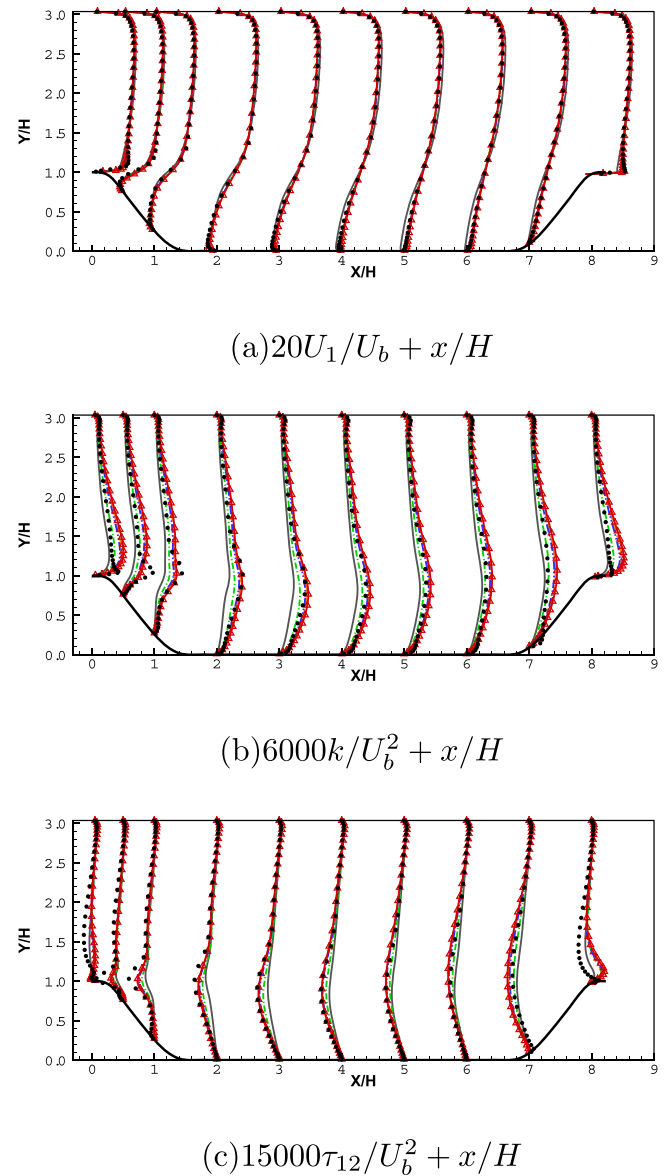
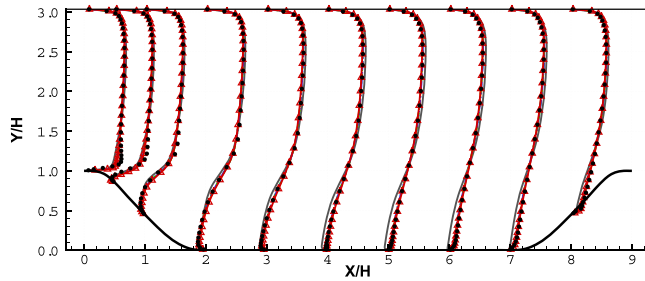
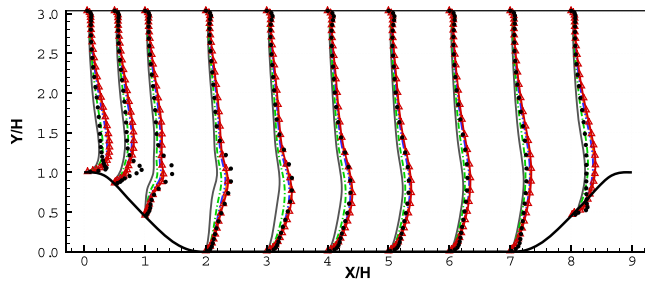


FIG. 13. Estimation for various SR models for flow over Periodic hills $Re = 5600$ and $\alpha = 0.8$.

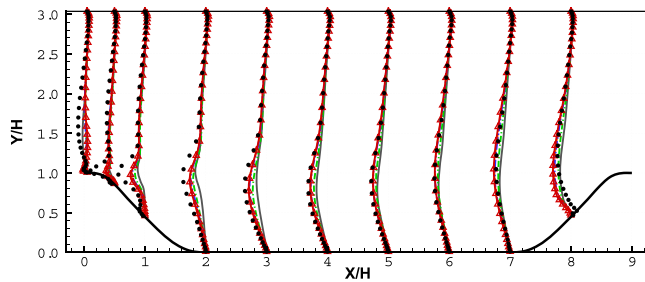
in specific classes of flows. Its success suggests the potential for extending this approach to a broader range of flow types and promoting the use of LLMs in various other fields of CFD. Utilizing the proposed framework based on natural language, human efforts can be greatly reduced. Designers are only required to concentrate on establishing appropriate evaluation criteria, where domain expertise is genuinely essential. A critical aspect of this approach is designing prompts that integrate the researcher's domain knowledge. These prompts guide the LLM in executing evolutionary algorithms, enabling it to incrementally improve solutions within the population and ultimately enhancing the quality and interpretability of the derived models.



(a) $20U_1/U_b + x/H$

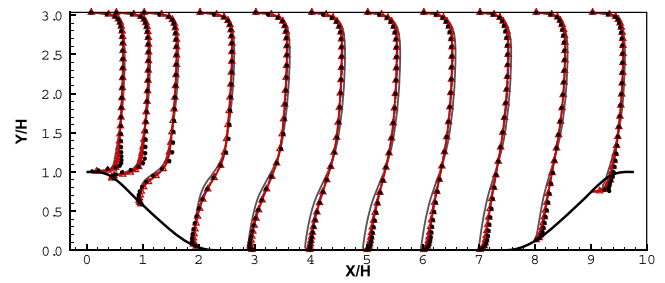


(b) $6000k/U_b^2 + x/H$

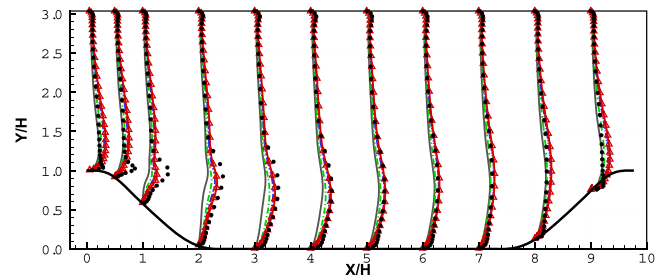


(c) $15000\tau_{12}/U_b^2 + x/H$

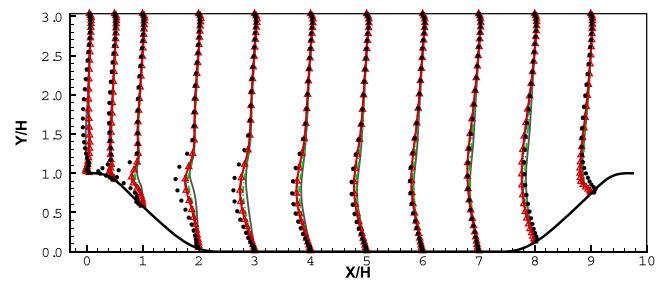
FIG. 14. Estimation for various SR models for flow over Periodic hills $Re = 5600$ and $\alpha = 1.0$.



(a) $20U_1/U_b + x/H$



(b) $6000k/U_b^2 + x/H$



(c) $15000\tau_{12}/U_b^2 + x/H$

FIG. 15. Estimation for various SR models for flow over Periodic hills $Re = 5600$ and $\alpha = 1.2$.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yu Zhang: Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Software (equal); Validation (equal); Writing – original draft (equal). **Kefeng Zheng:** Methodology (equal); Writing – original draft (equal). **Fei Liu:** Conceptualization (equal); Methodology (equal); Software (equal); Writing – original draft (equal). **Qingfu Zhang:** Supervision (equal); Writing – review & editing (equal). **Zhenkun Wang:** Supervision (equal); Writing – review & editing (equal).

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

APPENDIX A: PROMPTS FOR AUTOTURB

Initialization Prompt (IP)

Given two input variables I1 and I2, please help me design two non-linear expressions to approximate real-world equations.

The format must be as follows: <start> bDelta_ = T1*(...) + T2*(...) + T3*(...); RDelta_ = (T1*(...) + T2*(...) + T3*(...)) && tgradU()*2.0*k_<end> In this expression, replace '...' with appropriate functions of variables I1 and I2 and keep T1*, T2* and T3*. Do not modify && tgradU()*2.0*k_. Ensure the formulation is compatible with C++ programming standards.

Evolution Prompt (EP1)

Given two input variables I1 and I2, please help me design two non-linear expressions to approximate real-world equations.

I have 2 pairs of expressions as follows:

<Equation 1>

<Equation 2>

Please help me create a new expression that has a totally different form from the given ones.

The format must be as follows: <start> bDelta_ = T1*(...) + T2*(...) + T3*(...); RDelta_ = (T1*(...) + T2*(...) + T3*(...)) && tgradU()*2.0*k_<end> In this expression, replace '...' with appropriate functions of variables I1 and I2 and keep T1*, T2* and T3*. Do not modify && tgradU()*2.0*k_. Ensure the formulation is compatible with C++ programming standards.

Evolution Prompt (EP2)

Given two input variables I1 and I2, please help me design two non-linear expressions to approximate real-world equations.

I have 2 pairs of expressions as follows:

<Equation 1>

<Equation 2>

Please help me create a new expression that is motivated by the given ones.

The format must be as follows: <start> bDelta_ = T1*(...) + T2*(...) + T3*(...); RDelta_ = (T1*(...) + T2*(...) + T3*(...)) && tgradU()*2.0*k_<end> In this expression, replace '...' with appropriate functions of variables I1 and I2 and keep T1*, T2* and T3*. Do not modify && tgradU()*2.0*k_. Ensure the formulation is compatible with C++ programming standards.

Evolution Prompt (EP3)

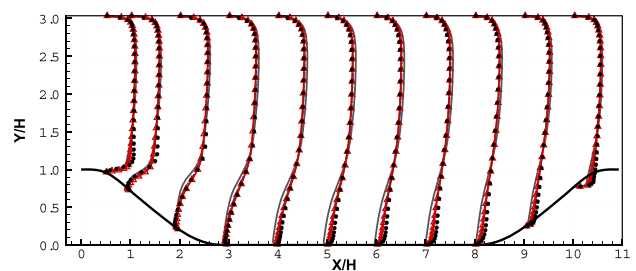
Given two input variables I1 and I2, please help me design two non-linear expressions to approximate real-world equations.

I have one expression as follows:

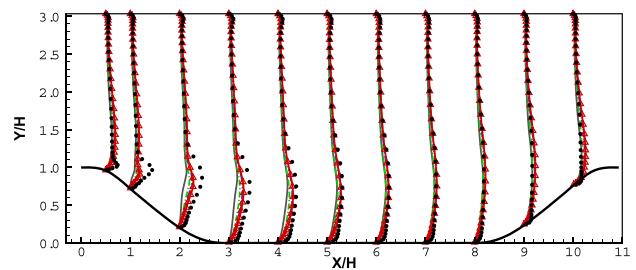
<Equation 1>

Please help me create a new equation that is a revision of the given one.

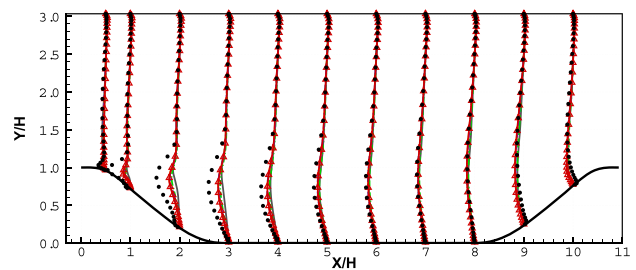
The format must be as follows: <start> bDelta_ = T1*(...) + T2*(...) + T3*(...); RDelta_ = (T1*(...) + T2*(...) + T3*(...)) && tgradU()*2.0*k_<end> In this expression, replace '...' with appropriate functions of variables I1 and I2 and keep T1*, T2* and T3*. Do not modify && tgradU()*2.0*k_. Ensure the formulation is compatible with C++ programming standards.



(a) $20U_1/U_b + x/H$



(b) $6000k/U_b^2 + x/H$



(c) $15000\tau_{12}/U_b^2 + x/H$

FIG. 16. Estimation for various SR models for flow over Periodic hills $Re = 5600$ and $\alpha = 1.5$.

Evolution Prompt (EP4)

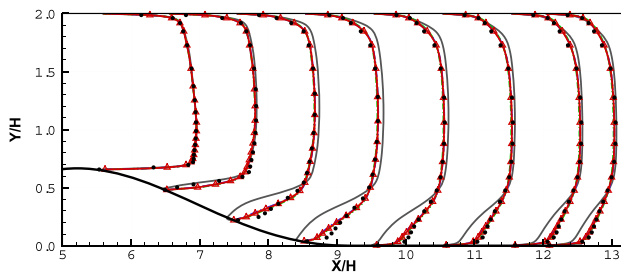
Given two input variables I1 and I2, please help me design two non-linear expressions to approximate real-world equations.

I have one expression as follows:

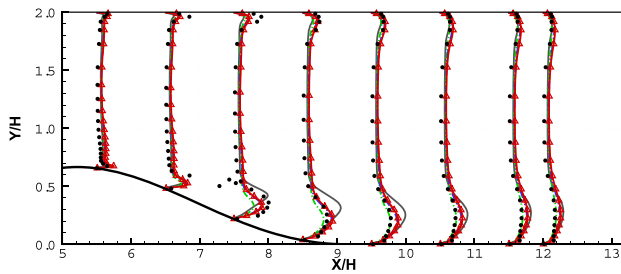
<Equation 1>

Please help me create a new equation that has different parameter settings of the given one.

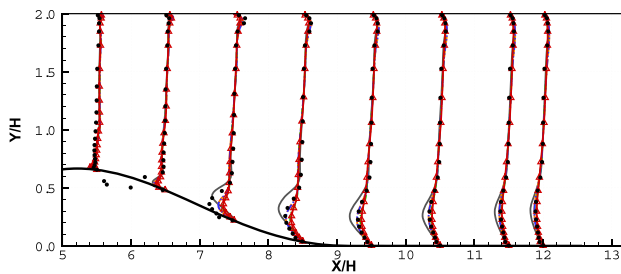
The format must be as follows: <start> bDelta_ = T1*(...) + T2*(...) + T3*(...); RDelta_ = (T1*(...) + T2*(...) + T3*(...)) && tgradU()*2.0*k_; <end> In this expression, replace '...' with appropriate functions of variables I1 and I2 and keep T1*, T2* and T3*. Do not modify && tgradU()*2.0*k_. Ensure the formulation is compatible with C++ programming standards.



(a) $U_1/U_b + x/H$



(b) $12k/U_b^2 + x/H$



(c) $25\tau_{12}/U_b^2 + x/H$

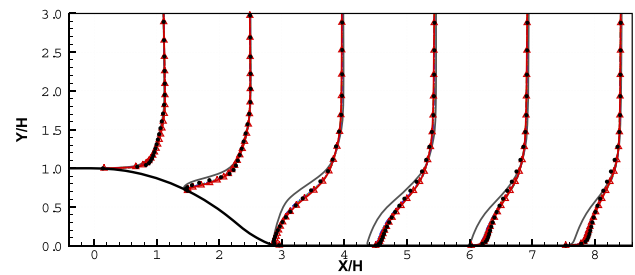
FIG. 17. Estimation for various SR models for flow over converging-diverging channel $Re = 12\,600$.

APPENDIX B: MEAN SQUARED ERROR OF PREDICTED FLOW QUANTITIES BY VARIOUS MODELS

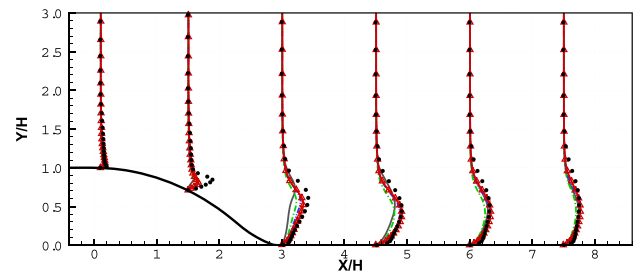
Mean squared errors of predictions for the mean velocity field U , turbulent kinetic energy k , and Reynolds shear stress anisotropy τ_{12} by various SR models for all the test cases: $PH\alpha_{0.8}$, $PH\alpha_{1.0}$, $PH\alpha_{1.2}$, $PH\alpha_{1.5}$, CD, CBF. In all cases, the errors are normalized with the MSE of the baseline $k-\omega$ SST model.

APPENDIX C: PREDICTED FLOW FIELDS BY VARIOUS SR MODELS FOR CROSS-VALIDATION FLOWS

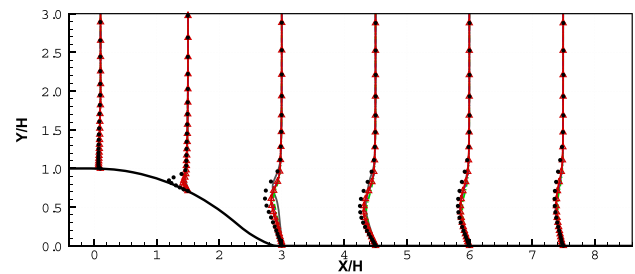
The profiles of U_1 , k , and τ_{12} at a sequence of streamwise locations for $PH\alpha_{0.8}$, $PH\alpha_{1.0}$, $PH\alpha_{1.2}$, $PH\alpha_{1.5}$, CD, CBF using various SR models.



(a) $U_1/U_b + x/H$



(b) $12k/U_b^2 + x/H$



(c) $20\tau_{12}/U_b^2 + x/H$

FIG. 18. Estimation for various SR models for flow over curved back-facing step $Re = 13\,700$.

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